

Predicting Auto Claim Count and Severity with GLMs

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Introduction

Which factors best predict claim frequency, severity, and total claim cost for auto insurance? The findings will be important for determining accurate and competitive premiums, directly impacting business profitability. The objectives are to develop GLMs to predict claim frequency and severity, evaluate their performances, and analyze the factors affecting each.

Methodology

Data

A dataset containing 20,000 policies was provided, each uniquely identified by ID. A second dataset containing only vehicle information was joined to the first dataset on vehicle make, body, and age after eliminating duplicate rows; this adds vehicle value as a predictor. A third dataset containing exposures was then joined by ID to the main dataset; this will be used as offsets to model frequency, where the response variable is claim count. For modeling severity, only 1,358 policies from the dataset with nonzero claim count were included. Severity (the response variable) is defined as total cost divided by claim count. The predictors under consideration for both models are:

- Driver Age Category (1-6)
- Gender (M,F) *corrected miscoded values
- Area (A-F)
- Vehicle make (A-D)
- Vehicle body (13 categories)
- Vehicle Age (1-4)
- Vehicle Value (continuous)

No missing data or errors were detected.

GLM Specification

A Poisson distribution was selected for claim counts; it is a discrete quantity, and the sample mean is close to the sample variance, indicating no significant overdispersion. A gamma distribution was selected for severity, due to its extreme right-skewed shape and lower bound at zero. To reduce the influence of extreme values on model fitting and validation, the top 3 outliers were removed, and the remaining losses were judgmentally capped at 12,250. A log link was used for both models, following the standard practice of using multiplicative rating in insurance.

One-way analysis

For each categorical predictor, the sample size, average claim count, and average severity was plotted for every level. Gender appears to contribute very little to risk differentiation. Vehicle body has several levels with sparse data; this presents practical difficulties for model fitting, thus avoiding it is preferred. Since the predictors are mostly categorical, detecting multicollinearity is more cumbersome. For simplicity, no serious correlations and interactions are assumed to exist among predictors; this assumption may lead to less reliable estimates and misleading interpretations.

Model Selection

Forward stepwise regression was performed using the entire dataset to guide variable selection; the model with the lowest AIC was chosen, effectively balancing goodness of fit with model complexity. The frequency model includes driver age, area, vehicle make and value as predictors; the severity model includes driver age, area, vehicle make and age. From a business perspective, this represents a varied mix of driver, territorial, and vehicle characteristics.

Variable Transformations

Uneven counts were prevalent among categorical levels, and coefficient stability was a major concern. Some variables were judgmentally transformed based on patterns observed from one-way analysis and stepwise regression (Base levels, selected to be the most populous group, are underlined).

- Driver age was coded as a numeric variable, as it displayed a monotonic relationship with the responses; this reduces the number of coefficients to be estimated.
- Area was grouped into {ABC,DEF} and {A, Other} for the frequency and severity model, respectively.
- Vehicle make was grouped into {A,B,CD} and {AB,CD} for the frequency and severity model, respectively.
- Vehicle age was grouped into {(1,2),(3,4)} for the severity model.

These groupings attempt to balance homogeneity and credibility, while also excluding insignificant effects (i.e. high p-values). Although this approach helps mitigate variance, it forfeits finer segmentation of risks, and may bias the predictions due to underfitting.

Model Validation

A 80/20 random train-test split was performed on the frequency model. Due to the lower volume of data and presence of extreme values, 10-fold cross validation was performed on the severity model to help reduce variance. Although there are some reversals present, the lift charts (Exhibits A,B) show that the frequency model strongly distinguishes risks; the highest and lowest decile averages differ by roughly 28 $\times$, whereas it is only 2 $\times$ for the severity model. The Lorenz curves (Exhibits C,D) corroborate these observations; high claim count predictions are concentrated in a smaller percentage of policies; while severity is much closer to the line of equality. Both diagnostics show the severity model performing especially poorly in the top decile.

Results and Conclusion

The models were re-fit on the entire dataset before being used to predict claim counts and severity for 10,000 new policyholders (See Exhibit E for model coefficients).

- Frequency and severity are influenced by similar factors, with driver age being the most stable predictor, having the narrowest CI in both models. However, its coefficient is rather small, suggesting a weak impact on prediction. It can be dropped from the model if collecting this data becomes inconvenient.
- The area and vehicle predictors have larger coefficients, indicating a stronger effect. Conversely, their CIs are wider; this variability may be driven by uneven levels or unidentified correlations. Although these factors enhance predictive power, it is recommended to apply credibility weighting or further monitor them over future periods to ascertain their underlying relationship with other variables.
- The levels are grouped differently in each model, reflecting subtle differences in the key drivers. For instance, vehicle make 'A' increases the expected claim count, whereas it decreases the expected severity.
- The frequency model predicts expected claim count, not actual claim count (which is a discrete value). If the model predicts an claim count of 0.15, then for 100 similar policies with the same exposure, 15 claims are expected. A policy-level actual-versus-predicted comparison would not be appropriate to evaluate the model.
- The severity model exhibits significant bias at extreme ranges, therefore human judgment is recommended for tail risks. These models are also useful for predicting total costs, defined as the product of claim count and severity.
- An indicator response variable was included in the test output to indicate whether a policy will have at least one claim. It is 1 if the predicted claim count is in the top 6.79% percentile, which is also the observed proportion in the training data.

Appendix

Exhibit A: Severity Model Lift Chart (left)

Exhibit B: Frequency Model Lift Chart (right)

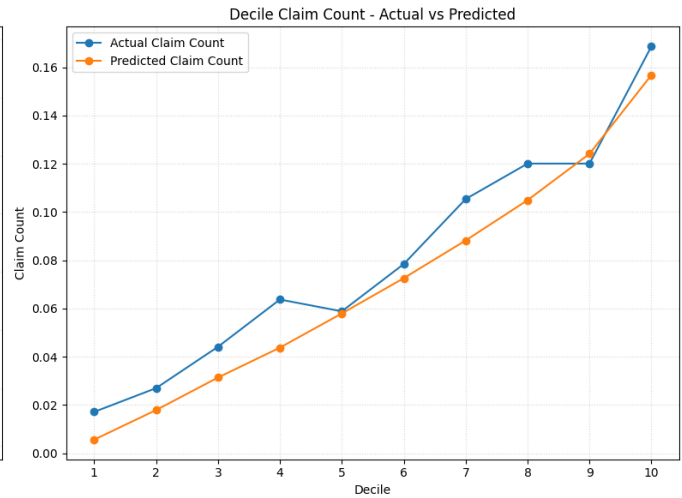
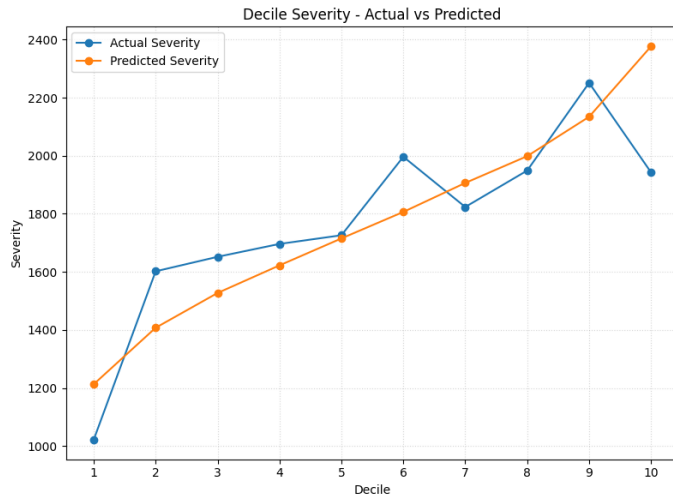


Exhibit C: Severity Model Lorenz Curve (left)

Exhibit D: Frequency Model Lorenz Curve (right)

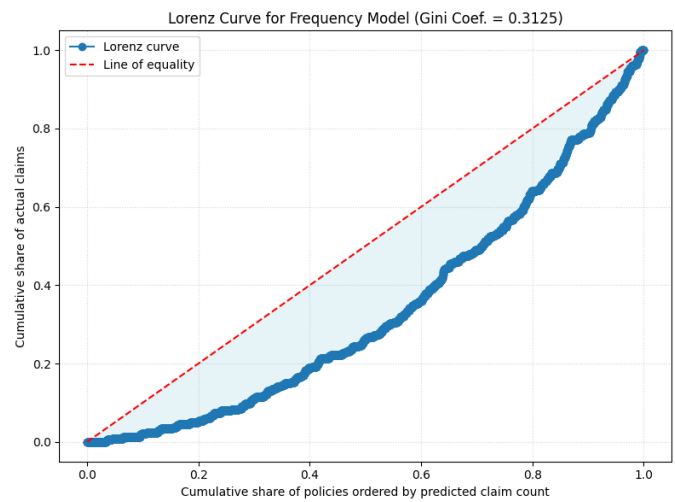
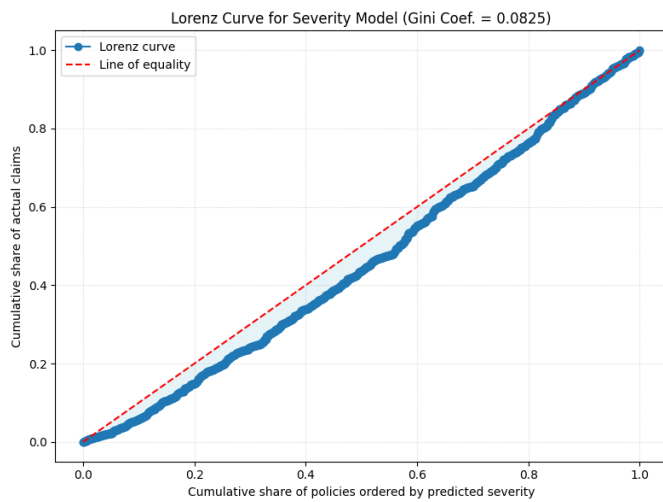


Exhibit E: Coefficient Estimates and Diagnostics

Frequency Model			
Predictor	Coefficient	95% CI	p-value
Intercept	-1.5980	[-1.737,-1.423]	0.000
Area (DEF)	-0.1368	[-0.225,0.015]	0.088
Vehicle Make (A)	0.4271	[0.058,0.779]	0.023
Vehicle Make (B)	-0.1110	[-0.206,0.046]	0.213
Vehicle Value	0.0952	[0.020,0.170]	0.013
Driver Age	-0.0982	[-0.134,-0.062]	0.000
Severity Model			
Predictor	Coefficient	95% CI	p-value
Intercept	7.9149	[7.708,8.122]	0.000
Area (A)	-0.2326	[-0.403,-0.062]	0.007
Vehicle Make (AB)	-0.1427	[-0.317,0.032]	0.109
Vehicle Age (1,2)	-0.1603	[-0.310,-0.010]	0.037
Driver Age	-0.0879	[-0.140,-0.036]	0.001